

CMOS 45nm Device TCAD Simulation

Executive Summary

This project presents a comprehensive TCAD simulation of a complementary metal-oxide-semiconductor (CMOS) device at 45nm technology node using Silvaco Atlas. The simulation characterizes DC performance including drain current modulation by gate and drain voltages, threshold voltage extraction, and device physics validation. Key results include extraction of critical parameters such as transconductance, subthreshold swing, and leakage currents. This project demonstrates proficiency in semiconductor device physics, TCAD methodology, and numerical simulation of advanced CMOS technology.

Motivation & Objectives

Motivation

CMOS technology remains the foundation of modern microelectronics, requiring precise device modeling to predict circuit behavior before fabrication. Device simulation using TCAD is essential for design optimization, technology scaling, physics validation, and educational understanding. The 45nm node represents a transition point where classical drift-diffusion physics begins to show limitations, making it ideal for studying modern MOSFET design challenges.

Objectives

1. Build and validate a 2D CMOS device structure using Silvaco Atlas 2. Extract DC characteristics (I_D vs V_{GS} , I_D vs V_{DS}) 3. Analyze device physics: threshold voltage, transconductance, subthreshold swing 4. Validate simulation against literature values for 45nm CMOS 5. Demonstrate TCAD workflow for reproducible device engineering

Device Architecture & Design

Physical Device Structure

Device Overview:

Property	Value
Technology Node	45 nm (educational model)
Configuration	Dual-gate CMOS (inverted twin-gate)
Substrate	Silicon bulk
Dielectric	SiO ₂ (oxide gate dielectric)
Gate Material	Aluminum

Key Design Parameters:

Parameter	Value	Unit	Rationale
Gate Length (L_g)	45 nm	nm	Defines technology node
Gate Oxide Thickness (T_{ox})	~10 nm	nm	Typical for 45nm node
Channel Width (W)	1.0 μm	μm	2D cross-sectional view
S/D Doping (N_D)	1×10^{20}	cm^{-3}	Heavy doping for low R_c
Substrate Doping (N_A)	3×10^{18}	cm^{-3}	Controls V_T

Simulation Methodology

TCAD Tool & Physics Models

Software: Silvaco Atlas TCAD (2D device simulator)

Physics Models Enabled:

Model	Purpose	Effect
Bipolar	Parasitic BJT effects	Impacts leakage at high V _{GS}
Fermi-Dirac	Quantum statistics	Important at high doping (S/D)
SRH	Recombination-generation	Dominant leakage in bulk
fldmob	Field-dependent mobility	Mobility degradation near surface
conmob	Concentration-dependent mobility	Velocity saturation in S/D

Numerical Solution & Mesh Strategy

Solver: Newton-Raphson with trapping (automatic convergence control)

Max iterations: 20 | Max trap iterations: 20

Mesh Resolution:

- Fine mesh near gate/channel: 0.002 μm (2 nm) spacing
- Coarse mesh in bulk: 0.01-0.02 μm (10-20 nm) spacing
- Total mesh elements: ~50,000-100,000
- Rationale: High resolution near gate for accurate field calculation

Results & Performance Metrics

DC Characteristics (I_D vs V_{GS} at $V_{DS} = 3V$)

Metric	Value	Unit	Notes
Threshold Voltage (V_T)	~0.4-0.5	V	Voltage where current turns on
Peak Transconductance (g_m)	~350-400	$\mu S/\mu m$	dI_D/dV_{GS} in saturation
Sub-threshold Swing (SS)	~75-90	mV/decade	Slope in weak inversion
On-state Current (I_{ON})	~1.3	mA	At $V_{GS} = V_{DS} = 3V$
Off-state Current (I_{OFF})	~10-100	pA	At $V_{GS} = 0V$, $V_{DS} = 3V$
I_{ON} / I_{OFF} Ratio	~ 10^7 - 10^8	—	Circuit switching capability

Output Characteristics (I_D vs V_{DS} at $V_{GS} = 3V$)

Metric	Value	Unit	Notes
Linear Region I_D	$I_D \propto V_{DS}$	mA	Ohmic region (0-0.5V)
Saturation Current (I_{sat})	~1.2-1.3	mA	Maximum drain current (>1.5V)
Early Voltage (V_A)	~50-100	V	Channel length modulation factor
Output Resistance (r_o)	~100-200	k Ω	dV_{DS} / dI_D in saturation
Drain Conductance (g_{ds})	~5-10	μS	Output impedance

Comparison with Published Literature

Parameter	This Work	Literature (45nm)	Match?
V_T (V)	0.4-0.5	0.35-0.5	✓ Yes
I_{ON}/I_{OFF}	10^7 - 10^8	10^6 - 10^7	✓ Good
SS (mV/dec)	75-90	70-90	✓ Good
g_m ($\mu S/\mu m$)	390	300-400	✓ Yes

Conclusion: Simulation results align well with 45nm CMOS expectations from ITRS projections and published literature.

Analysis & Discussion

Operating Regimes Observed

Sub-threshold Region ($V_{GS} < 0.4V$)

Physics: Weak inversion, diffusion-dominated | Current: $I_D \propto \exp(qV_{GS}/nkT)$ | $SS \approx 75-90$ mV/dec (excellent, near theoretical minimum)

Transition Region ($0.4V < V_{GS} < 1.5V$)

Physics: Moderate inversion | Current gain: $\sim 10-100\times$ per 100 mV increase in V_{GS} | Transition between diffusion and drift

Strong Inversion / Linear Region ($V_{GS} > 1.5V$, $V_{DS} < V_{GS} - V_T$)

Physics: Drift-dominated, full channel inversion | Model: $I_D = (W/L) \times \mu_n \times C_{ox} \times (V_{GS} - V_T) \times V_{DS}$ | I_D increases linearly with V_{DS}

Saturation Region ($V_{GS} > 1.5V$, $V_{DS} > V_{GS} - V_T$)

Physics: Channel pinch-off at drain end | Model: $I_D = \frac{1}{2} \times (W/L) \times \mu_n \times C_{ox} \times (V_{GS} - V_T)^2$ | Current nearly independent of V_{DS} (good current source)

Limitations & Physical Assumptions

2D Simulation Limitations

• 3D Effects Not Captured: Gate-induced drain leakage (GIDL), reverse short-channel effects, edge transistor behavior • Impact: Simulated I_{OFF} may underestimate real device • Solution: 3D TCAD would capture corner/edge effects

Physics Model Simplifications

• Doping Profile: Uniform vs. actual deep diffusion/implantation • Interface Charge: Idealized oxide interface (no fixed charge, trap states) • Quantum Effects: Drift-diffusion limited; no quantum tunneling models • Self-Heating: Temperature constant (no thermal analysis)

Temperature Effects

• Current Simulation: $T = 300K$ only • Real Operation: Devices reach 350-450K under circuit operation • Expected Changes: Mobility decreases $\sim 0.4\%/K$, V_T decreases ~ -1 mV/K, Overall I_D decreases with temperature

Reproducibility & How to Extend

Run This Simulation

1. Install: Silvaco Atlas (license required) 2. Navigate: `cd CMOS_45nm/silvaco_code` 3. Execute: `atlas -i cmos_45nm.in -o cmos.log` 4. Visualize: Open .str file in Tonyplot

Runtime Estimate: 10-15 minutes on standard workstation

Modify Parameters

Parameter	Original Code	Modified Example
Gate Length	<code>x.max=0.145 (45nm)</code>	<code>x.max=0.130 (30nm)</code>
Supply Voltage	<code>solve vdrain=3</code>	<code>solve vdrain=2</code>
Sweep Resolution	<code>vstep=0.1</code>	<code>vstep=0.05 (higher resolution)</code>

References

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Author: Ariful Islam Bhuiyan | Institution: CSUN | Advisor: Prof. Somnath Chattopadhyay
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